# Cardiac Deformation Lab

Comprehensive Research Analysis & Scientific Biomechanical Report

**Research Prototype — Computational Image Analysis**

Advanced continuum kinematics, dense optical-flow deformation fields, longitudinal metric kinetics, and mathematical validation benchmarks.

| Report Metadata Attribute | Specification |
|---|---|
| Study Name | Study Cohort Alpha - Phase 2 |
| Study ID | c8da97dc-ec26-4365-9787-4a4c74c41463 |
| Subject Code | SUBJ-A01 |
| Report Identifier | REP-EC667E9A |
| Generated Date | October 18, 2026 at 10:13:50 UTC |
| Software Version | v1.0.4 |
| Report Schema Version | v1.0 |
| Total Analyses Evaluated | 2 |

## MANDATORY SCIENTIFIC & NON-CLINICAL DISCLAIMER

*This document presents computational image-derived measurements generated by a research prototype. The results are not medical diagnoses, prognoses, treatment recommendations, or clinical findings.*
*Optical-flow displacement represents apparent image motion and may not equal physical tissue motion. All derived metrics (strains, directions, coherence, TSI) are exploratory mathematical descriptors.*

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## 1. Executive Summary

The study contains **2** computational image analyses evaluated under the research workspace pipeline. Of these, **0** are multi-frame cine temporal sequences, and **2** are differential single-pair analyses. Associated manual reference annotations count: **0**. Spatial physical calibration is active on **0** of 2 analyses, while localized Region-of-Interest (ROI) segmentation is enabled on **0** analyses.

| Metric | Value | Metric | Value |
|---|---|---|---|
| Total Image Analyses | 2 | Calibrated Analyses | 0 / 2 |
| Temporal Cine Sequences | 0 | ROI Segmentation Usage | 0 / 2 |
| Manual Reference Annotations | 0 | Average Computational QC Score | 96.0 / 100 |
| Total QC Technical Warnings | 0 | Longitudinal Tracking Available | Yes |

## 2. Study Information

| Study Attribute | Recorded Metadata |
|---|---|
| Study Name | Study Cohort Alpha - Phase 2 |
| Study Unique ID | c8da97dc-ec26-4365-9787-4a4c74c41463 |
| Subject Pseudonym | SUBJ-A01 |
| Creation Timestamp | October 18, 2026 at 10:13:48 UTC |
| Last Modified Timestamp | October 18, 2026 at 10:13:48 UTC |
| Study Description | Longitudinal cardiac motion study |
| Associated Analysis Sessions | 7a16acf5-7aa4-4645-8d61-83b8e5fbb5c7, e83a4c64-fe2a-40c0-a476-a23ac0dde30f |

## 3. Computational Methodology

The Cardiac Deformation Lab processes cardiac imagery through an automated, deterministic 12-stage computational continuum mechanics pipeline:
**Input** → **Preprocessing** (Grayscale conversion, Gaussian smoothing, histogram normalization) → **Motion Estimation** (Farnebäck dense optical flow) → **Displacement Calculation** (Euclidean vector magnitude) → **Spatial Strain Tensor** (Infinitesimal Green-Saint-Venant formulation) → **Principal Strains** (Spectral decomposition of eigenvalues) → **Advanced Kinematic Descriptors** (Motion coherence, strain gradient, anisotropy, deformation energy) → **Temporal Analysis** (Inter-frame cine tracking, TSI) → **ROI & Tracking** (Fixed or Lucas-Kanade updated contours) → **Computational Quality Control** (Signal SNR, gradient smoothness, flow bounds) → **Statistical & Longitudinal Analytics**.

## 4. Mathematical Framework

**Displacement Vector:**
**d**(x, y) = [u(x, y), v(x, y)]■
Displacement magnitude: $D = \sqrt{(u^2 + v^2)}$

**Displacement Gradient Tensor:**
$\nabla$**d** = [ [∂u/∂x, ∂u/∂y], [∂v/∂x, ∂v/∂y] ]

**Infinitesimal Strain Tensor:**
ε = 1/2 ($\nabla$**d** + ($\nabla$**d**)■)
ε_xx = ∂u/∂x
ε_yy = ∂v/∂y
ε_xy = 1/2 (∂u/∂y + ∂v/∂x)

**Principal Strains & Directions:**
Eigenvalues of ε: λ■, λ■ = ((ε_xx + ε_yy)/2) ± $\sqrt{(((\varepsilon_xx - \varepsilon_yy)/2)^2 + \varepsilon_xy^2)}$

**Radial and Circumferential Deformation:**
d_r = **d** · **e**_r (projection onto unit radial vector from centroid)
d_c = **d** · **e**_c (projection onto unit tangential vector)

**Spatial Calibration:**
Physical displacement = pixel displacement × pixel spacing (mm/px).
*Note: Strain represents dimensionless spatial gradient (mm/mm) and remains invariant to isotropic scaling.*

## 5. Input and Preprocessing

| Analysis ID | Input Image Pair | Dimensions | Grayscale | Normalization |
|---|---|---|---|---|
| 7a16acf5 | frame_1.png / frame_2.png | N/A | Complete | Normalized (8-bit) |
| e83a4c64 | frame_1.png / frame_2.png | N/A | Complete | Normalized (8-bit) |

## 6. Motion Analysis

| Analysis ID | Mean | Max | Min | Std Dev | Units | Method |
|---|---|---|---|---|---|---|
| 7a16acf5 | 2.1000 | 4.5000 | 0.0000 | 0.4000 | px | Farnebäck Dense Flow |

| Analysis ID | Mean | Max | Min | Std Dev | Units | Method |
|---|---|---|---|---|---|---|
| e83a4c64 | 2.4000 | 4.9000 | 0.0000 | 0.5000 | px | Farnebäck Dense Flow |

**Figure 1: Estimated Image Displacement Magnitude Field [Metric: Displacement] (Units: px)**

*[Visual file not found at outputs/disp.png]*

***Caption:*** *Figure: Estimated image displacement magnitude computed via dense optical flow.*

## 7. Strain Analysis

| Analysis ID | Mean Exx | Max Exx | Mean Eyy | Max Eyy | Mean Exy | Mean ε■ | Max ε■ |
|---|---|---|---|---|---|---|---|
| 7a16acf5 | 0.0500 | N/A | -0.0400 | N/A | 0.0100 | 0.0900 | 0.1800 |
| e83a4c64 | 0.0600 | N/A | -0.0500 | N/A | 0.0100 | 0.1100 | 0.2000 |

## 8. Advanced Kinematic Metrics

| Analysis ID | Motion Coherence | Strain Gradient | Anisotropy | Strain Entropy | Deform Energy |
|---|---|---|---|---|---|
| 7a16acf5 | 0.8800 | N/A | 0.3500 | N/A | 12.4000 |
| e83a4c64 | 0.8900 | N/A | 0.3800 | N/A | 14.2000 |

## 9. Computational Deformation Fingerprint

| Dimension Key | Biomechanical Descriptor | Raw Scalar | Normalized Value [0-1] |
|---|---|---|---|
| D_mean | Mean Displacement | 2.1000 | N/A |
| D_max | Max Displacement | N/A | N/A |
| epsilon_mean | Mean Strain Magnitude | N/A | N/A |
| epsilon_max | Peak Strain Magnitude | N/A | N/A |
| epsilon_principal | Principal Strain (ε■) | 0.0900 | N/A |
| anisotropy | Strain Anisotropy | N/A | N/A |
| entropy | Strain Entropy | N/A | N/A |
| strain_gradient | Strain Spatial Gradient | N/A | N/A |
| coherence | Motion Coherence Index | N/A | N/A |
| TSI | Temporal Stability Index | N/A | N/A |

## 10. Temporal Cine Analysis

*Temporal analysis was not available for this study.*

## 11. Region of Interest (ROI) and Tracking Analysis

**Important Scientific Protocol:** Full-image metrics and localized ROI metrics are evaluated separately. They are never mixed or aggregated into a single indiscriminate field.

*No localized ROI segmentation was applied to the analyses in this study.*

## 12. Spatial Calibration

**Calibration Status:** Displacement remains in image-space pixels.
*Displacement values are converted to physical units only when valid spatial calibration is available.*

## 13. Computational Quality Assessment

***Notice:*** *This quality assessment evaluates computational/image-processing conditions and is not a clinical confidence score.*

| Analysis ID | Quality Score | Rating Level | Warnings | Primary Warning Message |
|---|---|---|---|---|
| 7a16acf5 | 96.0 / 100 | High | 0 | N/A |
| e83a4c64 | N/A / 100 | N/A | 0 | N/A |

## 14. Longitudinal Multi-Session Analysis

| Metric Name | Trend Direction | Slope | $R^2$ | Abs Change | Pct Change |
|---|---|---|---|---|---|
| Mean Displacement | stable | N/A | N/A | N/A | N/A |
| Maximum Displacement | stable | N/A | N/A | N/A | N/A |
| Mean Strain | stable | N/A | N/A | N/A | N/A |
| Maximum Strain | stable | N/A | N/A | N/A | N/A |
| Mean Principal Strain | stable | N/A | N/A | N/A | N/A |
| Maximum Principal Strain | stable | N/A | N/A | N/A | N/A |
| Anisotropy Index | stable | N/A | N/A | N/A | N/A |
| Strain Entropy | stable | N/A | N/A | N/A | N/A |
| Strain Gradient | stable | N/A | N/A | N/A | N/A |
| Motion Coherence | stable | N/A | N/A | N/A | N/A |
| Temporal Stability Index | stable | N/A | N/A | N/A | N/A |
| ROI Mean Displacement | stable | N/A | N/A | N/A | N/A |
| ROI Mean Strain | stable | N/A | N/A | N/A | N/A |
| ROI Principal Strain | stable | N/A | N/A | N/A | N/A |
| Computational Quality Score | stable | N/A | N/A | N/A | N/A |

## 15. Advanced Statistical Analytics

| Metric | N | Mean | Median | Std Dev | Min | Max | IQR |
|---|---|---|---|---|---|---|---|
| Mean Displacement | 2 | 2.2500 | 2.2500 | N/A | 2.1000 | 2.4000 | 0.1500 |
| Max Displacement | 2 | 4.7000 | 4.7000 | N/A | 4.5000 | 4.9000 | 0.2000 |
| Mean Strain | 2 | 0.0550 | 0.0550 | N/A | 0.0500 | 0.0600 | 0.0050 |
| Max Strain | 0 | N/A | N/A | N/A | N/A | N/A | N/A |
| Mean Principal Strain | 2 | 0.1000 | 0.1000 | N/A | 0.0900 | 0.1100 | 0.0100 |
| Max Principal Strain | 2 | 0.1900 | 0.1900 | N/A | 0.1800 | 0.2000 | 0.0100 |
| Anisotropy Index | 2 | 0.3650 | 0.3650 | N/A | 0.3500 | 0.3800 | 0.0150 |
| Strain Entropy | 0 | N/A | N/A | N/A | N/A | N/A | N/A |
| Strain Gradient | 0 | N/A | N/A | N/A | N/A | N/A | N/A |
| Motion Coherence | 2 | 0.8850 | 0.8850 | N/A | 0.8800 | 0.8900 | 0.0050 |
| Temporal Stability Index | 0 | N/A | N/A | N/A | N/A | N/A | N/A |
| Quality Score | 1 | 96.0000 | 96.0000 | N/A | 96.0000 | 96.0000 | 0.0000 |

*Notice: Correlation describes statistical association and does not imply causation.*

## 16. Synthetic / Phantom Validation Benchmark

***Methodological Context:*** *Synthetic validation evaluates algorithmic behavior against mathematically controlled reference fields and does not establish clinical validity.*

| Parameter | Setting | Validation Error Metric | Observed Value |
|---|---|---|---|
| Deformation Mode | N/A | Displacement MAE | N/A px |
| Deformation Strength | N/A | Displacement RMSE | N/A px |
| Noise Amplitude | N/A | Displacement Pearson r | N/A |
| Gaussian Blur Sigma | N/A | Principal Strain MAE | N/A |

## 17. Manual Reference Benchmark

***Important Notice:*** *Manual annotations are reference measurements for computational benchmarking and are subject to annotator variability. They are not clinical ground truth.*

*No manual landmark or contour annotations available for reference benchmarking in this study.*

## 18. In-Study Comparison Analysis

*Differential comparison could not be calculated.*

## 19. Limitations

**1. Apparent Image Motion vs. Physical Motion:** Optical flow measures brightness constancy variations in image pixel coordinates, which may differ from physical tissue motion under out-of-plane speckle decorrelation or non-rigid myocardial through-plane deformation.
**2. Two-Dimensional Restriction:** All strain tensors (ε_xx, ε_yy, ε_xy) are computed in a 2D image plane, omitting radial-longitudinal shear components present in 3D cardiac mechanics.
**3. Temporal Sampling Rate:** Frame rates affect peak displacement and strain derivative estimation. Nyquist limits govern kinematic precision.
**4. Spatial Calibration Constraints:** Physical conversions (mm/px) assume isotropic, uniform pixel spacing in orthogonal directions.
**5. Manual Annotation Variability:** Reference annotations represent subjective operator placements with intra- and inter-observer variance.
**6. Statistical Sample Size:** Small study cohorts limit statistical generalizability; correlations represent association rather than causation.

## 20. Computational Conclusion

The analysis demonstrates the extraction of image-derived motion, displacement, strain, and related computational descriptors using the implemented research pipeline. The pipeline successfully processed all associated input frames, computed dense continuum fields, evaluated spatial and temporal consistency, and compiled quantitative multi-session trends.

**Scientific Interpretation Notice:** The results should be interpreted as computational research outputs and not as clinical findings. These measurements provide an objective quantitative basis for engineering research, computational algorithm validation, and academic inquiry.

## 21. Appendix & System Metadata

| System Component | Technical Specification |
|---|---|
| Software System | Cardiac Deformation Lab (CDL) |
| Software Version | v1.0.4 |
| Report Engine | ReportLab PDF Engine with Dynamic Two-Pass Canvas |
| Report Identifier | REP-EC667E9A |
| Timestamp of Generation | October 18, 2026 at 10:13:50 UTC |
| Optical Flow Algorithm | Farnebäck Dense Optical Flow (OpenCV) |
| Strain Formulation | Infinitesimal Green-Saint-Venant Deformation Tensor |
| Included Analysis Sessions | 7a16acf5, e83a4c64 |